

Table 1: Results of MindTS compared with unimodal and LLM-based methods on six real-world datasets. These methods only use the time series in the dataset. The best results are highlighted in bold, and the second-best results are underlined.

Datasets	Metric	MindTS	DADA	LMixer	UTime	Timer	UniTS	G4TS	Modern	TsNet	DC	A.T.	Patch	TranAD	iTrans	PCA	IF	LODA	HBOS
Weather	Aff-F	82.66	69.01	73.68	76.46	75.46	76.17	72.56	81.06	80.58	42.80	49.22	77.17	77.81	75.37	64.91	54.06	52.55	47.70
	V-PR	57.48	30.00	43.47	51.90	43.21	44.35	41.30	52.13	50.09	18.33	19.17	50.03	52.08	42.56	47.19	49.66	<u>55.03</u>	46.58
	V-ROC	82.64	61.03	71.71	78.45	73.22	75.08	70.03	81.14	<u>81.91</u>	45.56	43.32	79.97	78.75	73.37	57.38	56.45	57.00	54.16
Energy	Aff-F	74.37	64.38	65.85	61.98	60.20	63.84	66.37	70.76	66.00	47.07	43.39	66.85	49.69	70.81	57.65	62.03	63.45	55.85
	V-PR	50.36	34.18	30.35	32.88	29.46	31.04	31.68	36.60	38.61	22.57	19.69	34.41	33.80	35.82	44.30	46.03	48.63	42.57
	V-ROC	74.44	54.37	53.04	49.97	46.03	51.15	53.10	<u>65.05</u>	59.47	45.93	31.56	58.31	56.37	63.06	53.07	53.61	55.90	51.50
Environment	Aff-F	85.29	84.11	<u>84.36</u>	81.71	84.19	83.06	72.26	81.07	80.41	62.24	59.75	81.17	61.41	74.43	55.63	46.50	46.45	22.25
	V-PR	56.79	<u>54.20</u>	52.94	48.87	51.42	50.24	23.94	42.26	50.64	7.69	18.14	45.78	4.91	24.87	17.87	8.94	18.66	52.15
	V-ROC	93.78	87.69	89.75	91.77	<u>92.10</u>	92.03	66.79	89.78	87.97	41.28	51.98	90.86	14.20	73.81	37.08	46.20	50.69	51.03
KR	Aff-F	90.28	84.22	71.80	88.58	<u>89.55</u>	82.24	79.56	84.42	85.47	61.94	70.99	79.52	73.26	79.49	58.11	69.38	60.96	64.78
	V-PR	53.15	45.90	15.13	46.87	51.41	43.32	38.23	39.95	51.60	8.48	7.94	36.18	28.42	27.37	24.19	43.31	51.82	<u>52.06</u>
	V-ROC	89.86	70.82	52.79	73.55	75.99	67.81	<u>88.87</u>	79.00	43.04	41.97	74.65	64.18	41.05	76.12	47.51	60.70	59.99	61.41
EWJ	Aff-F	83.89	81.26	66.86	78.22	78.06	77.61	76.65	81.57	<u>81.82</u>	48.10	59.03	75.82	69.22	78.27	51.06	67.55	72.06	71.03
	V-PR	50.42	43.36	15.21	32.39	33.17	39.32	35.63	<u>44.75</u>	43.15	15.37	10.85	36.08	17.80	28.98	19.38	37.81	40.08	41.19
	V-ROC	84.12	71.79	46.80	64.49	67.72	73.91	67.95	<u>83.88</u>	75.76	47.10	31.75	71.56	49.60	72.16	45.26	59.24	61.65	62.07
MDT	Aff-F	89.19	77.99	67.65	76.28	78.51	75.57	<u>80.81</u>	80.08	47.33	66.12	79.47	63.93	78.66	54.66	53.74	55.06	52.33	
	V-PR	65.44	46.81	19.10	38.94	38.38	37.61	44.81	<u>82.18</u>	50.53	15.72	15.93	41.67	14.34	36.36	22.93	35.32	44.63	44.77
	V-ROC	83.02	66.76	47.06	61.00	60.28	58.67	62.30	<u>82.30</u>	79.56	45.02	44.53	77.69	28.55	71.87	44.09	54.02	55.98	55.30

Table 2: The notation with * indicates the results of extending the baselines to their multimodal versions using the recent time series multimodal framework MM-TSFLib, where both time series and text data from datasets are utilized.

Datasets	Metric	MindTS	DADA*	LMixer*	UTime*	Timer*	UniTS*	G4TS*	Modern*	TsNet*	Patch*	TranAD*	iTrans*
Weather	Aff-F	82.66	69.73	76.30	73.93	75.37	76.38	76.82	81.50	80.09	77.05	77.73	75.36
	V-PR	57.48	30.42	45.94	43.19	43.36	44.58	45.83	<u>53.42</u>	50.53	50.17	52.09	40.30
	V-ROC	82.64	61.51	75.29	73.71	73.26	75.21	74.61	81.67	<u>82.06</u>	80.08	78.72	70.11
Energy	Aff-F	74.37	64.80	61.46	65.38	60.36	65.42	67.38	72.13	66.71	66.28	50.53	72.49
	V-PR	50.36	34.38	30.91	30.44	29.57	31.34	31.83	37.44	<u>38.88</u>	34.66	33.74	36.21
	V-ROC	74.44	55.63	49.06	49.85	46.39	51.89	53.52	<u>66.37</u>	59.80	58.47	56.38	65.62
Environment	Aff-F	85.29	83.84	83.76	76.73	<u>84.52</u>	83.43	84.44	81.36	80.21	81.71	61.38	76.02
	V-PR	56.79	54.20	51.22	35.64	51.20	50.06	<u>56.65</u>	41.36	50.39	45.52	4.93	25.85
	V-ROC	93.78	88.02	91.74	84.20	<u>92.02</u>	91.98	90.22	89.14	88.14	90.87	14.29	73.66
KR	Aff-F	90.28	84.22	<u>90.03</u>	77.38	89.61	83.06	88.29	84.87	85.84	79.52	72.50	78.39
	V-PR	53.15	45.68	52.98	37.12	51.56	44.27	57.93	40.86	51.73	36.37	28.47	28.12
	V-ROC	89.86	72.08	75.21	66.96	75.92	74.25	80.43	<u>89.22</u>	78.94	74.80	41.24	77.17
EWJ	Aff-F	83.89	81.41	78.23	73.20	79.05	78.04	81.37	81.88	81.92	76.49	69.03	78.23
	V-PR	50.42	43.18	34.06	25.71	33.36	39.99	43.93	<u>45.41</u>	43.28	36.22	17.87	29.79
	V-ROC	84.12	71.06	69.37	67.57	68.19	74.39	76.93	<u>83.98</u>	75.91	72.21	49.85	74.11
MDT	Aff-F	89.19	77.89	78.37	72.33	77.93	76.70	81.86	81.68	80.62	78.87	63.60	77.47
	V-PR	65.44	47.22	42.61	25.31	38.23	37.78	<u>52.65</u>	45.88	52.30	41.85	14.55	33.88
	V-ROC	83.02	68.06	61.72	49.16	60.21	58.82	73.39	<u>82.66</u>	73.57	77.72	28.88	69.95